Fast and Accurate Mouse Brain MRI Registration with *VoxelMorph*

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***Abstract*—Image registration is essential for aligning anatomical structures in medical imaging, but traditional methods are computationally intensive and struggle with variability in mouse brain MRI. In this paper, we applied VoxelMorph, a deep learning-based framework for deformable image registration, to improve alignment accuracy and efficiency. We fine-tuned a pre-trained human brain model on a mouse brain MRI dataset and implemented preprocessing steps to standardize inputs and enhance anatomical consistency. The fine-tuned model outperformed the generic version, achieving lower similarity losses, fewer deformation artifacts, and improved anatomical plausibility. This approach provides a scalable and accurate solution for mouse brain registration in preclinical research.**

***Keywords*—MRI, image registration, mouse brain, VoxelMorph, deep learning.**

I. Introduction

Image registration is the process of aligning two or more images to geometrically align datasets for comparison across different modalities, subjects, or even across time[1]. For this project, we will be focusing on registration of MRI images taken from mouse brains. Traditional mouse brain registration faces several limitations such as high dimensionality and resolution, which is particularly difficult to align when dealing with the nonlinear deformations inherent in biological tissues[2]. Another main challenge is the computational complexity involved in image alignment, since traditional registration methods optimize an objective function for each pair of images.

Deep learning techniques in medical image registration have made significant advances in improving accuracy and efficiency to create 3D models from 2D slices. Deep learning models consist of many sequential layers and parameters that allow them to learn complex information, to produce accurate predictions[4]. They do this by diversifying the training dataset which reduces overfitting and using pre-trained weights with optimization algorithms to enhance the model's ability to find global minima. They also increase processing speed during inference by avoiding iterative optimization with passing image pairs through a single forward pass[5]. These advantages make these models optimal for brain image registration.

Some techniques that have been widely used in the past 8 years include 1) DeepSlice, 2) DeepMapi, and 3) Bi-Channel Image Registration and Deep-Learning Segmentation (BIRDS). DeepSlice is a Convolutional Neural Network (CNN) trained to register mouse brain images to the Allen Brain Common Coordinate Framework[6]. DeepMapi is a CNN that predicts the deformation field for each pair of images and automatically registers micro-optical imaging datasets to a reference atlas[7]. BIRDS is a platform that combines registration with neural networks consisting of image preprocessing, bi-channel registration (raw image data and feature map), and automatic annotation. This method can segment incomplete brain data following registration since its network extracts pixel-level features[8].

Our proposed model, VoxelMorph[3], is a CNN that maps input image pairs and computes deformation fields to align the images. This method has proven to have an increased computational speed and can be applied to large datasets by training with a regularization term that ensures smooth deformations. It is also transferable to other anatomies besides the brain. The biggest limitation is the need for high-quality images to provide an accurate registration. Our study aims to use VoxelMorph as an optimization tool for improving registration accuracy across mouse brain MRI images.

We aim to develop a deep learning framework for mouse brain MRI registration using VoxelMorph[3], a deep learning model for deformable image registration. First and foremost we aim to train a VoxelMorph model to learn spatial transformations for precise alignment of MRI scans. In addition, we hope to adapt the model to handle variations in image quality including differences in contrasts and resolutions or other possible batch effects. We then hope to optimize the model for computational efficiency to provide a scalable and efficient tool to aid in the analysis of structural and functional mouse brain changes.

II. Methods

To create the VoxelMorph based model for our project, we used the following steps detailed below.

A. Dataset

The dataset consisted of 61 T2 RARE mouse brain MRI subject scans (moving) and 1 template scan (fixed). The

VoxelMorph model we used was pre-trained using human brain 3D T1 MRI scans.

B. Preprocessing

To prepare the input MRI scans for registration using VoxelMorph, we applied the following preprocessing pipeline to ensure anatomical consistency and standardize our inputs across images.

1. *N4 Bias Correction.* Since MRI images often have scanner artifacts that cause image intensity to be non-homogenous, we applied the ants.n4_bias_field_correction function to our images. This helped to reduce low-frequency intensity variations across the images and helped enhance tissue contrast.
2. *Skull Stripping.* We used skull stripping to remove all of the non-brain tissue from our MRI scans. This improves our registration by focusing the alignment on the actual brain structure and eliminating irrelevant features. This was done using ANTsPY.
3. *Affine Alignment:* We affine aligned our scans to a common reference space to remove any large-scale spatial differences across subjects. This helps bring all of the images into rough alignment before we input to VoxelMorph to learn the fine-grained deformation fields.
4. *Intensity Normalization.* We then normalized images to have 0 mean and 1 variance to ensure consistency across subject scans.
5. *Resampling.* Finally, we resampled our images to 160 x 192 x 224 to match the input shape of the pretrained model.

C. VoxelMorph Architecture

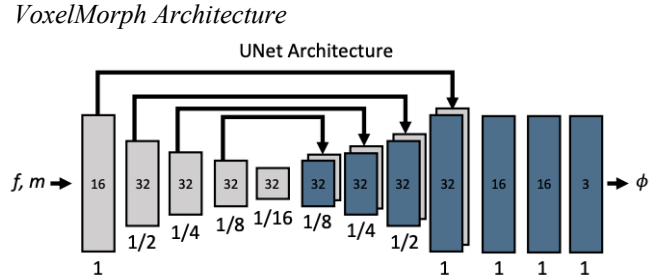

Fig. 1. VoxelMorph's architecture is a 3D convolutional U-Net that maps a pair of images to a dense displacement field. Encoder layers progressively reduce spatial resolution to extract features, while decoder layers use 32-filter convolutions and upsampling to reconstruct the full-resolution field. Final convolutions refine the output at full resolution.

We used VoxelMorph (Fig. 1), an unsupervised deep learning framework for deformable image registration. Its architecture consists of a U-Net-style encoder-decoder that takes a concatenated pair of moving and fixed 3D images and outputs a dense displacement field. This field is then applied to the moving image using a spatial transformer to generate the registered output. The model is trained using a combination of image similarity loss and a smoothness regularization term to enforce plausible deformations.

D. Experimental Setup

For training, the loss function was defined as: $L(f, m, \varphi) = L_{sim}(f, m \circ \varphi) + \lambda L_{smooth}(\varphi)$. Where L_{sim} is $MSE = \frac{1}{|\Omega|}\sum_{p\in\Omega} [f(p) - [m \circ \varphi](p)]^2$. The similarity loss function penalizes differences in appearance between the moved and fixed images, while the smoothness loss function penalizes non-smooth deformations. The L2 smoothing regularization parameter λ=0.01 was added to prevent overfitting. Additionally, we trained the model with 20-40 epochs and an Adam optimizer with a learning rate of 1e-4.

III. Results

To evaluate the performance of the VoxelMorph-based image registration framework, we compared a pre-trained model trained on generic human brain MRI data with a fine-tuned model trained on the mouse dataset. The results highlight how fine-tuning improves spatial alignment for domain-specific anatomical structures.

A. Loss Function Determination for Training

To evaluate the most effective similarity metric for this task, we compared the fine-tuned model performance with three loss functions: Mean Squared Error (MSE), Normal Cross-Correlation (NCC), and Similarity Index Measure (SSIM). Each loss function was evaluated based on its impact on visual alignment and deformation quality (Fig. 2). MSE, a voxel-wise intensity-based metric, offered stable convergence and sharp alignment in high-contrast regions. NCC, being invariant to linear intensity differences, performed slightly better in globally aligning lower-contrast structures but introduced more spatial warping. SSIM, which accounts for local structure and contrast, produced visually smooth results but was more sensitive to noise in some scans. Ultimately, MSE was selected for both the pretrained and fine-tuned models due to its consistent performance across subjects and its compatibility with gradient-based regularization.

A large number of voxels with non-positive Jacobian values indicates undesirable folding or compression in the deformation field. Table I. displays the number and percentage of voxels with non-positive Jacobian determinants. MSE had 0.01% Jacobians, suggesting a good preservation of the original anatomy. NCC had slightly more Jacobians, 0.81%, showing a good performance but SSIM had the higher number of Jacobians, 3.6%, indicating an unnatural anatomy deformation. MSE was chosen as the best loss function to train the models.

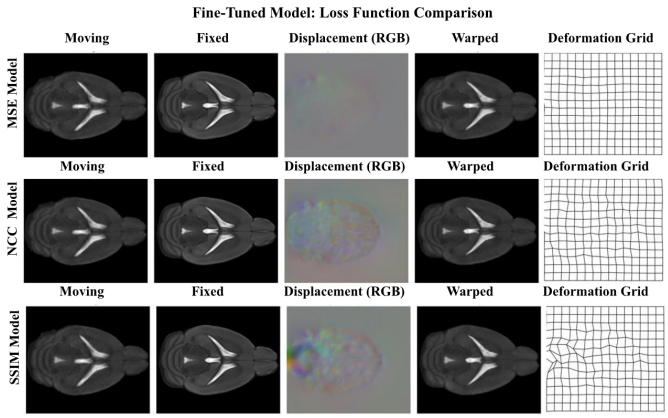

Fig. 2. Comparison between similarity loss functions MSE, NCC, and SSIM. The alignment, displacement fields (φ), warping, and deformation grid are shown for each function.

TABLE I. FINE-TUNED MODEL: LOSS FUNCTION COMPARISON

Loss Function	Jacobian Matrix (|Jφ|) ≤ 0	% Jacobian Matrix (|Jφ|) ≤ 0
MSE	74	0.01%
NCC	6,950	0.81%
SSIM	31,145	3.62%

B. Comparison Between Fine-Tuned and Pre-Trained Models

A qualitative assessment was performed by randomly selecting test subjects and visualizing the warped outputs from both models (Fig.3 and Fig.4). The fine-tuned model produced warped images with notably better anatomical alignment to the fixed template, particularly in smaller brain regions where shape variability is higher. The deformation fields, visualized via RGB-coded displacement maps, appeared smoother and more localized in the fine-tuned model across multiple samples.

A quantitative comparison was performed between the pretrained and fine-tuned models by computing the mean, standard deviation (STD), and standard error of the mean (SEM) of voxel-wise similarity metrics across a test set. For each registered pair, the voxel-wise MSE between the warped moving image and the fixed image was calculated. The fine-tuned model consistently achieved lower MSE values, indicating improved alignment. Aggregated across all test subjects, the fine-tuned model had a lower average error (mean ± SEM), demonstrating both improved accuracy and reduced variability. The standard deviation was also smaller for the fine-tuned model, suggesting more consistent performance across anatomical variations. These results quantitatively reinforce the visual improvements observed in the warped outputs (Fig. 5).

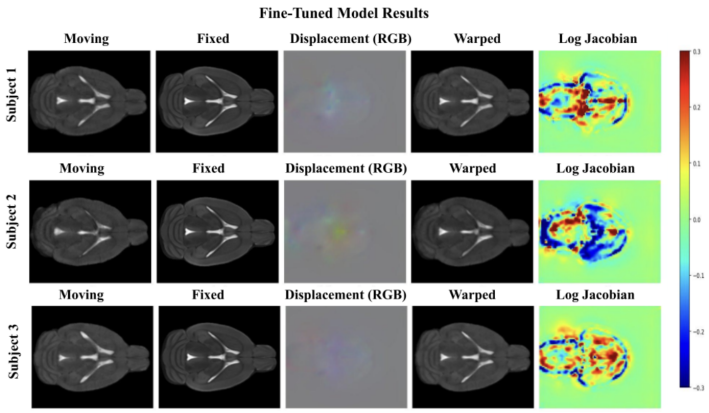

Fig. 3. Fine-tuned Model results for three different subjects.

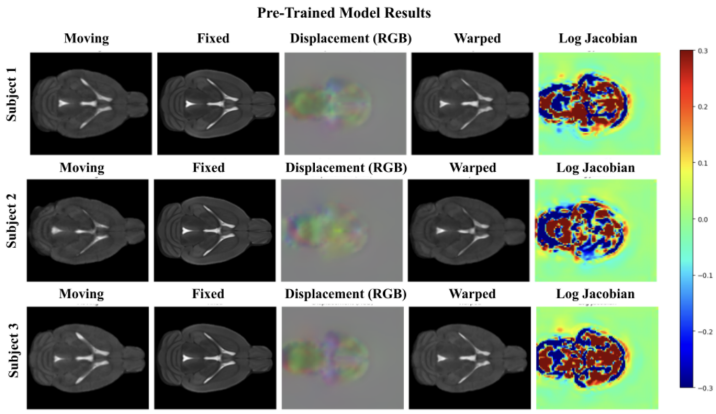

Fig. 4. Pre-trained Model results for three different subjects.

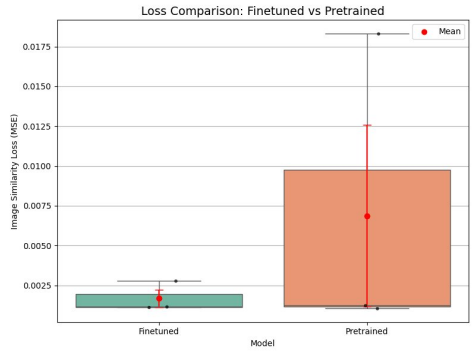

Fig. 5. MSE similarity loss comparison (mean and standard deviation) between the pre-trained and fine-tuned models.

Finally, to quantify the anatomical plausibility of the deformation fields, we computed the Jacobian determinant of the spatial transforms which is summarized in Table II. The pre-trained model exhibited a higher number of voxels with non-positive Jacobian determinants, while the fine-tuned model reduced these artifacts substantially. For instance, the percentage of voxels with non-positive Jacobian determinants decreased from 2.73% (pre-trained) to 0.09% (fine-tuned),

indicating that fine-tuning resulted in more anatomically realistic deformations.

TABLE II. PRE-TRAINED VS. FINE-TUNED MODELS: MSE LOSS FUNCTION COMPARISON

Model	Mean	STD	SEM	$\lvert J_\varphi \rvert \leq 0$	% $\lvert J_\varphi \rvert \leq 0$
Pre-Trained	0.006870	0.009903	0.005717	23,515	2.73%
Fine-Tuned	0.001691	0.000940	0.000940	780	0.09%

IV. CONCLUSIONS

In this paper, we addressed the challenge of accurate and efficient medical image registration, a fundamental task in medical imaging that traditionally relies on computationally intensive optimization-based methods. VoxelMorph presents a solution to this problem by leveraging deep learning to perform fast, unsupervised image registration, which reduces inference time while maintaining accuracy. It is also flexible enough to apply to other anatomies and imaging modalities such as CT and multimodal scans. However, despite its advantages, VoxelMorph has limitations, including sensitivity to training data diversity and difficulty in handling large deformations or anatomical abnormalities not well-represented in the training set. Additionally, the reliance on a smoothness prior may oversimplify complex anatomical changes. To mitigate some of these limitations, we implemented preprocessing steps to improve registration accuracy and robustness.

Our results demonstrate that fine-tuning a pretrained VoxelMorph model on mouse brain MRI data significantly improves alignment quality and anatomical plausibility. Compared to the pre-trained model, the fine-tuned version yielded lower similarity losses, fewer non-positive Jacobian voxels, and more consistent performance across subjects. This indicates that domain-specific adaptation is essential for achieving optimal performance in non-human or specialized datasets.

Future work should focus on adding anatomical segmentation maps during training to improve structural fidelity. There is also a need to adapt models for registration of pathological brains, where anatomical landmarks may be distorted. Lastly, exploring lightweight models or hybrid approaches could make learning-based registration more accessible in clinical environments with limited computational resources.

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CODE

GitHub Repository:
https://github.com/vivianlu63/VoxelMorph.git